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Semi-supervised anomaly detection algorithms: A comparative summary and future research directions

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ABSTRACT

While anomaly detection is relatively well-studied, it remains a topic of ongoing interest and challenge, as our society becomes increasingly interconnected and digitalized. In this paper, we focus on existing anomaly detection approaches, by empirically studying the performance of 29 semi-supervised anomaly detection algorithms on 95 benchmark imbalanced databases from the KEEL repository. These include well-established and commonly used classifiers (e.g., One-Class Support Vector Machine (ocSVM) and Isolation Forest) and recent proposals (e.g., BRM and XGBOD). Findings from our in-depth empirical study show that BRM is a robust classifier, in terms of achieving better classification results than the other 28 state-of-the-art techniques on diverse anomaly detection problems. We also observe that OCKRA, Isolation Forest, and ocSVM achieve good performance overall AUC, but poor classification results on databases where the number of objects is equal or greater than 1,460, all features are nominal, or the imbalance ratio is equal or greater than 39.14.

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1. Introduction

Anomaly detection aims to find patterns in a database whose behavior is not normal or expected [1–3]. Such unexpected behaviors are generally referred to as anomalies. As a learning task, anomaly detection may be supervised, semi-supervised, or unsupervised [4–6]. A key challenge is the lack of quality training instances. For example, it is generally easier to get access to labeled instances for normal behavior, but getting access to labeled set(s) of anomalous instances is challenging, particularly for newer systems and device types (e.g., Internet of Things (IoT)) [7]. In addition, different types of anomalies (that did not appear in the training phase) can appear in the test phase [8]. In other words, anomaly detection remains challenging [9,10], especially in areas where it is difficult to obtain a well-sampled positive class. One example is in the development of medical diagnostic systems [11] or industrial fault detectors [12]. Measurements of typical conditions (negative objects), say in airplane turbines, are easy to collect; however, failures (positive objects) require an error to occur or recording the occurence at the exact moment, which is generally not viable.

We posit the potential of semi-supervised anomaly detection (SSAD) in overcoming these challenges. Hence, in this paper, we focus on SSAD, where the goal is to build a model for the class corresponding to normal behavior, and use the model to identify anomalies in the test data. There exist several SSAD algorithms proposed in the literature, such as ocSVM (One-Class Support Vector Machine) [13], LOF (Local Outlier Factor) [14], kNN (k-Nearest Neighbor) [15], and AutoEncoders (AE) [16]. While several of these techniques achieve good predictive performance, new SSAD algorithms are continually being introduced to solve specific problems or for specific systems. The increasing number of SSAD methods and publicly available-benchmark databases [17–20] also motivates this research. Specifically, in this paper we seek to conduct in-depth empirical performance analysis for the different detection algorithms.

A number of researchers have empirically compared different (older) SSAD methods in multiple databases, but the evaluation is generally limited to a small number of databases and to a specific domain. For example, the study carried out by Janssens et al. compared anomaly detection methods in the fields of Machine Learning and Knowledge Discovery on several databases. The

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authors only reported the performance of four methods, including SVDD (Support Vector Data Description) [22], Parzen Window Density Estimation [23], LOF (Local Outlier Factor) [14], and LOCI (Local Correlation Integral) [24].

However, by focusing research on a particular problem, newer classifiers such as MO-GAAL (Multiple-Objective Generative Adversarial Active Learning) [20], FastABOD (Fast Angle-Based Outlier Detection) [25], and OCKRA (One-Class K-means with randomly projected features algorithm) [19] are not usually included in these comparisons; despite (potentially) obtaining a higher classification performance than conventional (or older) detection algorithms.

Selecting suitable classifiers for the experimental setup is not always an easy task. Usually, authors select well-established classifiers (e.g., ocSVM [13]) or those that follow the same approach (i.e., based on clustering, nearest neighbor, and/or statistical techniques), ignoring recent anomaly detection proposals that have shown good classification results (e.g., BRM [26]).

We also observe a large number of literature review/survey articles on anomaly detection in the literature. For example, Chandola et al. provided a structured overview of anomaly detection techniques and application domains. Alam et al. discussed the advantages and disadvantages of several ocSVM-based techniques. Taha and Hadi surveyed 36 different methods for anomaly detection on categorical data and showed the similarities and differences between these methods. Hence, this is not the scope of this paper. Rather, as discussed above, we evaluate the performance of 29 state-of-the-art SSAD algorithms, including popular SSAD algorithms (e.g., ocSVM and Isolation Forest) and newer algorithms like MO-GAAL or XGBOD (Extreme Boosting Based Outlier Detection) [28] (both proposed in the last two years), on 95 benchmark (imbalanced) databases to determine which of them are the top performers. Furthermore, for corroborating our results, we execute a statistical comparison to analyze the differences among the performance of the classifiers fairly. We also provide an in-depth study of the tested detection methods regarding the different used databases' features, such as the total number of objects and features, type of features, and imbalanced ratio (IR). To the best of our knowledge, our comparison is the first attempt to provide a broad empirical study in terms of the number of SSAD methods and databases, which are not related to a specific application domain.

The rest of this paper is organized as follows. Section 2 presents an overview of SSAD approaches, including existing performance evaluations. Next, Section 3 briefly reviews existing state-of-the-art SSAD approaches. In Section 4, we describe our experimental setup and the empirical evaluations. Finally, in Section 5, we present the conclusion and future research agenda.

2. Semi-supervised anomaly detection approaches

In this section, we introduce an overview of the SSAD topic. We also describe what the main characteristics presented in a problem to be dealt with as a SSAD are. Besides, we discuss how SSAD addresses class imbalance problems.

Anomaly detection aims to identify instances that come from a different distribution or class than the majority [4–6]. In contrast to the well-known supervised classification techniques where a training set containing objects from all classes is used to train a classifier to identify to which of a set of pre-defined classes an object belongs, anomaly detection depends on the labels available in the database. Typically anomalies in data are rare (or difficult to capture) compared to normal instances. In SSAD techniques, the training set has only labeled instances of the normal class and does not require labels for the anomaly class [4].

For example, consider the problem of masquerader detection [26], in which the model must determine whether the activity corresponds to the legitimate user or an intruder. For carrying out it, the model tries to identify any attempt to impersonate a legitimate user to gain unauthorized access to sensitive data or services for that user by comparing each access pattern regarding those patterns describing the accesses of the legitimate user [29].

For dealing with masquerader detection using a multi-class classifier, it requires several objects representing the legitimate user behavior (normal or negative class) and objects representing attacks made on the computer (anomaly or positive class). However, while negative objects are easy to collect, positive objects are more complicated for collecting [26]. In this way, while a multi-class classifier needs to find decision boundaries among all classes of the problem, an SSAD algorithm builds a model using only the negative class and tries to distinguish it from the positive class.

For a better illustration, Fig. 1 shows an example of the problem outlined above. Given a training set, we can represent each object o as a point in the feature space (two-dimensional, for this case). Red stars indicate the data representing the legitimate user class (negative class), and the impostor classes (positive class) are represented by blue and green circles (attack type 1 and 2, respectively). Notice that using multi-class classification can separate the problem's classes without error through boundaries among the labeled objects (see the dashed line). Nevertheless, only legitimate users and attacks of type 1 and type 2 can be detected by using multi-class classification. Also, in Fig. 1(a), notice that the trained multi-class model only can classify query objects in one of the three pre-defined classes; as a consequence, new types of attacks or anomalies could be classified in the legitimate user class (see query objects in Fig. 1(a), which were represented as purple squares). However, it can be solved by using a One-Class Classification (OCC) anomaly detection technique for setting a boundary around the negative class; in this way, the classifier can distinguish the patterns describing the legitimate user data from those patterns describing impostors or attacks. See in Fig. 1(b) the dashed line representing the OCC boundary and the query objects (purple squares).

There are different categories to group anomaly detection techniques and each one has unique assumptions regarding the nature of the anomalies made by the techniques in each category [4]. For example, OCC-based anomaly detection techniques try to define the decision boundaries by capturing the negative class' general behavior for accepting as many negative objects as possible and minimizing the possibility of admitting positive objects [18,30]. Section 3 will explain these categories in more detail. However, for the sake of simplicity, Fig. 1(b) illustrates an example where an OCC algorithm is used to solve an SSAD algorithm builds SSAD task.

While SSAD is useful when only data from a single class is available for training, it has also become a viable option when multi-class classification does not perform well on class imbalance problems [31]. Sometimes, the lack of objects representing the positive class is due to privacy issues, high measurement costs, or the low frequency with which abnormal events occur [32,33]. Consequently, there are significantly more objects belonging to the negative class than those belonging to the positive class, triggering inefficacy problems in multi-class classifiers because they are commonly biased by the negative class and ignore the positive class [34]. Anomaly detection problem can be considered as an extreme case of class imbalance problem [35] and a suitable solution is to apply SSAD techniques when the positive class is poorly sampled [36].

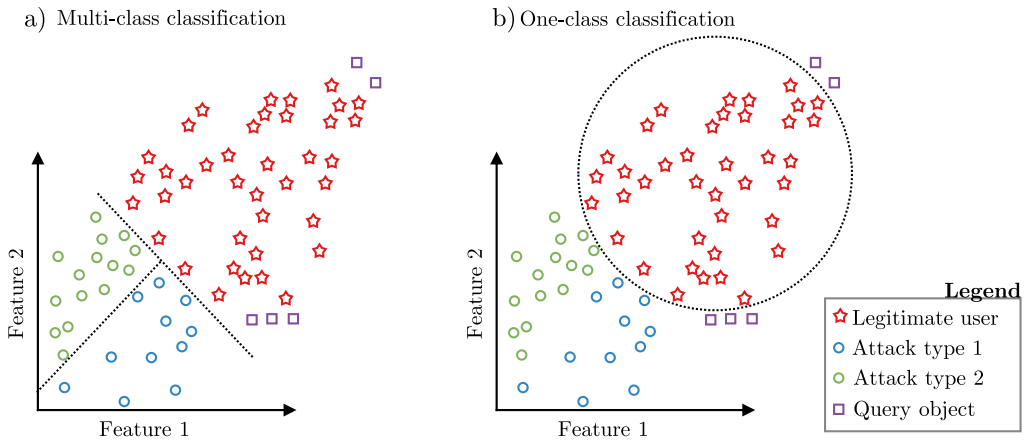


Fig. 1. Examples of decision boundaries of a multi-class and a one-class classifier. (a) The dashed lines represent the class boundaries, which distinguish among the three classes (green and blue circles and red stars). (b) The dashed line draws the decision boundary around the negative class; hence, using a one-class classifier can differentiate the legitimate user data from unknown attacks (query objects).

2.1. Applications of SSAD

SSAD has been used in several real-world applications. For all cases mentioned below, the goal is to find patterns describing positive and negative-query objects regarding those contained in the training data. We mention here just few examples:

- Fraud detection: We aim to identify criminal activity in credit card purchases, financial transactions, among others. To achieve this, data from previous transactions are analyzed. If any change (deviation) in behavior is observed, it is possible that it is a fraudulent transaction [37].
- Medical diagnostics: This field usually works with patient records to detect unusual symptoms or test results that indicate possible health problems. For example, this information can be used for breast cancer diagnosis [38].
- Industrial monitoring and damage detection: Anomaly detection methods provide useful information about machine performance and identify failures as soon as possible, to reduce maintenance and repair costs. The methods can be applied for machine fault detection [12], identification of anomalies in a gas turbine combustion system [39], among others.
- Social media: There are many applications that make use of anomaly detection in social media such as finding fake news [40], and automated rumor detection [41].
- Recommendation systems: These systems aim to improve the user experience through personalized recommendations based on implicit user feedback (e.g., [42]).
- IT Security: Anomaly detection methods are applied to identify malicious programs in the operating system [43], detect unauthorized access to computer systems o different behavior from that of normal users [44], verify that the right user is given access (e.g., speaker verification [45], signature verification [46], etc.), among others.
- Tex mining: Pattern recognition techniques facilitate the automatic identification of harmful conversations [47]. In real-world applications collecting normal conversations is more practical than collecting abnormal or harmful behavior.

2.2. Existing SSAD performance evaluations

Many efforts have been made to improve conventional SVDD to address the problem of anomaly detection. For example, Sadeghi and Hamidzadeh proposed Automatic SVDD (ASVDD) and conducted experiments on six benchmark databases taken from the UCI Machine Learning repository[1] [49]. The results demonstrate the superiority of the proposed method over state-of-the-art ones in terms of classification accuracy and AUC. Hamidzadeh and Moradi also compared their new proposal with three versions of SVDD, the Wilcoxon test proves the proposed method is significantly different from others regarding the accuracy, precision, and recall.

In [17], Désir et al. proposed a new classifier called One-Class Random Forests (OcRF), which was compared using 13 databases (transformed into anomaly detection, giving a total of 78 databases) and four reference techniques, such as Gaussian density model (GDM) [51], Parzen estimator [23], Gaussian Mixture Models (GMM) [52], and ocSVM [13]. According to the results, OcRF performed significantly better than Parzen and ocSVM, and in high-dimensional spaces, OcRF obtains stable predictions. However, GDM is the best of the five tested methods; although, OcRF is among the three most successful classifiers for three-quarters of the tested databases.

An important proposal was developed by Liu et al. where their authors assessed the performance of four classifiers (LOF [14], Random Forest (RF) [54], ORCA [55] –a k-Nearest Neighbor (kNN) based method–, and their proposed classifier, Isolation Forest or iForest) using 11 databases for anomaly detection. The advantage of their proposal is that it has low memory requirements, and it converges quickly in high-dimensional spaces. In terms of AUC and CPU time, iForest outperformed the other tested classifiers.

The study presented by Rodríguez et al. introduced an ensemble for personal risk detection. The proposed method, OCKRA, was compared against ocSVM and three versions of the Parzen window classifier [23], using several anomaly detection databases created by the authors. OCKRA achieved a significant statistical improvement over ocSVM and demonstrated that the combination of multiple anomaly detectors outperforms the performance obtained by a single one.

In the online signature verification context, Nanni presented an experimental study comparing seven classifiers (GDM [51], GMM [52], kNN [15], SVDD [22], Parzen [23], Principal component analysis (PCA) [57], and Linear programming description (LPD) [58]). Results obtained on the Subcorpus-100 MCYT signature database [59] showed that Parzen and PCA achieved the best performance.

To solve the problem of drug profiling in forensic science, in [60], the authors evaluated seven different classifiers (K-means,

[1] http://archive.ics.uci.edu/ml.

Table 1
Summary of previous works on SSAD methods comparison.

Reference	Classifiers	Databases	Evaluation criteria	Statistical comparison	Application domain	Best classifier(s)
[56]	GDM, GMM, kNN, PCA, SVDD, LPD, Parzen	Subcorpus-100 MCYT	EER	None	Online signature verification	Parzen and PCA
[60]	K-means, kNN, Parzen, two version of GMM, SVDD, GDM	Own database	ROC curves, AUC	None	Drug profiling	SVDD
[53]	iForest, LOF, RF, ORCA	11 databases (Http and Smtp of KDD CUP 99, Pima, Breast cancer, Satellite, etc.)	AUC, CPU time	None	New algorithm proposed	iForest
[21]	LOF, LOCI, kNN, Parzen, SVDD	24 UCI databases (Arrhythmia, Hepatitis, Diabetes, Survival, etc.)	AUC	Friedman test with Nemenyi post-hoc, CD diagram	Experimental comparison	SVDD and LOF
[17]	OcRF, GDM, Parzen, GMM, ocSVM	78 UCI databases (Iris, Diabetes, Breast cancer, Pendigits, etc.)	Accuracy, MCC	Friedman test with Nemenyi post-hoc	New algorithm proposed	OcRF
[48]	ASVDD and three versions of SVDD	6 UCI databases (Breast cancer, Glass, Iris, Seeds, etc.)	Accuracy, AUC	Wilcoxon's signed-rank test	New algorithm proposed	ASVDD
[61]	ocSVM, three versions of Parzen using Mahalanobis distance	PRIDE dataset	P-R curves, ROC curves, AUC, EAR, Accuracy	Friedman test with Bergmann-Hommel post-hoc, CD diagram	Personal risk detection	ocSVM, but there is no significant difference with the rest of them.
[19]	OCKRA, ocSVM, three versions of Parzen using Euclidean distance	PRIDE dataset	P-R curves, ROC curves, AUC	Wilcoxon's signed-rank test	New algorithm proposed for personal risk detection	OCKRA
[50]	Proposal and three versions of SVDD	17 UCI databases (Diabetes, Yeast Vehicle, Nursery, etc.)	Precision, Recall, FPR, Accuracy, AUC	Wilcoxon's signed-rank test	New algorithm proposed	The new method
[62]	Naïve Bayes, ocSVM, OCKRA, BTPM, BRM	1 genuine accounts database and 3 spambots databases	AUC	Friedman test with Bergmann-Hommel and Schaffer post-hocs, CD diagram	Twitter bot detection	There is no statistical difference between the four classifiers. BTPM is the best-ranked algorithm.
[63]	ODIN, K-means, FastABOD, ocSVM, LOF, HBOS	4 intrusion detection databases (KDD Cup 99, NSL-KDD, ISCX2012, UNSW-NB15) and 3 synthetic databases	MCC, TP, FP, TN, FN, Precision, Recall, FPR, Accuracy, AUC	None	Intrusion detection	LOF, ODIN and ocSVM

kNN, Parzen, two versions of GMM, SVDD, and GDM). In most cases, SVDD seemed to be more robust, even though GDM and kNN obtained the highest AUCs.

In the area of personal risk detection, ocSVM was compared against Parzen estimator and two versions of Parzen based on K-means, all of them using the Mahalanobis distance [64]. Using a custom repository, called PRIDE [61], of anomaly detection databases, the authors found that ocSVM has remarkable results, and their performance is significantly better than the other three tested methods [61].

For bot detection in Twitter, Rodríguez-Ruiz et al. analyzed the performance of five classifiers that had been used so far for masquerade detection and personal risk detection, such as OCKRA [19], Bagging-TPMiner (BTPM) [65], and Bagging-RandomMiner[2] (BRM) [26]. They also included ocSVM and Naïve Bayes in that comparison. The results by Rodríguez-Ruiz et al. showed that BTPM was the best-ranked classifier, followed by OCKRA. However, there was no significant statistical difference in the compared classifiers.

Janssens et al. published a study comparing five different anomaly detection algorithms using 24 different databases. The authors concluded that SVDD [22] and LOF [14] performed best, and the results obtained by LOCI [24] and Parzen [23] were significantly worse than the rest of the tested classifiers.

Besides studies evaluating conventional algorithms only, there are also a few articles that study anomaly detection in data stream (see, e.g., [66,67], and [68]). However, in this paper, we do not evaluate anomaly detection methods taking into account the streaming data, so we do not include such methods.

Finally, Table 1 presents a summary and a brief comparison of the classifiers selected in the reviewed papers. As can be noted, not many comparative studies on SSAD do exist today. Most studies are focused on evaluating state-of-the-art anomaly detection methods in specific domains. Those cases that introduce a new algorithm compare against a few well-known anomaly detection techniques. Moreover, these works' outcomes are not always comparable, as different databases and evaluation metrics are used. Hence, it is essential to perform an up-to-date comparison of several SSAD methods on many benchmark databases to help researchers to see a fair comparison.

3. State-of-the-art SSAD approaches

In this section, we analyzed several state-of-the-art SSAD approaches designed for different approaches such as linear (Section 3.1), proximity (Section 3.2), probabilistic (Section 3.3), neural networks (Section 3.4), and ensembles (Section 3.5). The main

[2] BRM can be installed as a python package at https://pypi.org/project/brmi ner.

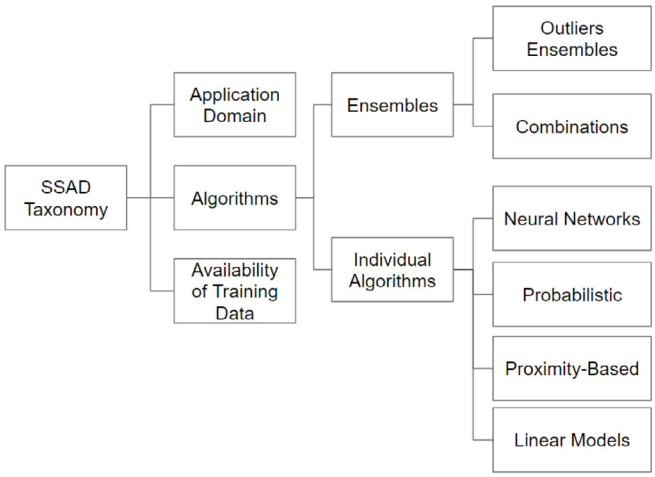


Fig. 2. Extended SSAD Dtaxonomy based on the proposal of Khan and Madden.

idea is to show an up-to-date review for each one of these one-class classification approaches.

SSAD is a well-known Machine Learning problem that has attracted much attention from the Machine Learning community [69,70]. Currently, there exist several taxonomies, but for our up-to-date comparison, we propose to analyze the most outstanding one-class classifiers by using our extended taxonomy, which is based on the one proposed by Khan and Madden [36]. Fig. 2 shows our extended taxonomy of SSAD.

Algorithms following a SSAD approaches can be divided into two categories: (a) individual and (b) ensembles. On the one hand, classifiers following the individual category are grouped into four different clusters: (1) Linear models, (2) Proximity-based, (3) Probabilistic, and (4) Neural Networks. On the other hand, classifiers following the ensembles category are fusing some of these four clusters above. Below, we detailed each category's characteristics, as mentioned earlier, and the most outstanding SSAD classifiers [36,69,70].

3.1. Linear models

Linear models comprise a linear combination of features. Linear models have had a good performance for different domains, including semi-supervised anomaly detection methods. For example, the Minimum Covariance Determinant (MCD) [71,72] applies the Mahalanobis distance [64] in a matrix by using a consistent high-decomposition multivariate location. MCD finds outliers in a cluster setting using an F distribution.

Another representative method widely used to SSAD classification is principal component analysis (PCA) [57]. The PCA method has been part of SSAD classifiers designed for detecting anomalies. In a problem with anomalies, a predictive intrusion model is constructed from the major and minor principal components of the negative class. A measure of the difference of an anomaly from the negative class is the distance in the principal component space [57].

One-Class Support Vector Machines (ocSVM) is a classifier that moves towards the OCC and SSAD by attempting to compute a function F. A kernel enlargement gives the functional form of F in terms of a small subset of the training data. It is regularized by controlling the extent of the vector in a feature space [73,74].

3.2. Proximity-based

For many real-world applications, finding anomalies can be more exciting than finding ordinary (usual) behavior. Existing works in anomaly detection consider it an outlier detection problem [14,75]. In both, the problem depends on the degree of how isolated the object is concerning the surrounding neighborhood. A prominent approach for anomaly detection is the proximity-based [14,76].

One of the first proximity-based approaches for OCC and ASSAD classifiers is the Local Outlier Factor (LOF) [14]. LOF recognizes anomalies by comparing the objects with their neighbors. Those objects belonging to the normal (or negative) class and objects belonging to anomaly (positive) class are recognized depending on their neighbors [77]. LOF computes the distance among objects for discovering the anomalies. As a result, LOF performs better when data contains few features and many objects. Usually, LOF presents a stable performance, and their classification results are reliable and accurate [78].

In [79], Tang et al. proposed a variant of LOF, named Connectivity-based Outlier Factor (COF), which enhances the effectiveness of the original LOF. On the other hand, He et al. proposed a cluster-based local outlier detection algorithm, which provides importance to the local data behavior. In this work, the authors proposed a measure for computing the significance of an outlier, called cluster-based local outlier factor (CBLOF) [80,81].

k-NN [82] is a classifier widely used in OCC and SSAD [15,76]. k-NN starts from the assumption that an object belongs to the same class as its nearby objects [83]. k-NN uses feature similarity for discovering the generality describing the objects in the training database, which further means that query objects will be assigned to the class of their nearest neighbors; usually, the majority vote scheme is the most used.

3.3. Probabilistic

The results obtained by probabilistic-based models depend on chance; they are also sometimes called randomized models [84]. Probabilistic-based models have been applied to anomaly detection, showing good classification results [25,85–87].

In [25], Kriegel et al. proposed ABOD (Angle-Based Outlier Detection), a SSAD classifier for detecting anomalies in large databases. ABOD is not as sensitive to distance since it measures the difference among the angles of an object (abstracted as a vector) to the other objects. In this way, the dimensionality is alleviated compared to other approaches computing the distance between objects [85,88]. The main trouble of ABOD is its time complexity, $O(n^3)$, which is too high. As a consequence, FastABOD was proposed by the same authors [25]. FastABOD is an alternative based on a sample of the database for improving its time complexity [89]. In FastABOD, objects far away from an object o is assumed to be less important than objects nearby of o. In this way, the fast angle-based outlier factor was proposed to use pairs of objects by taking into account its k-nearest neighbors.

In [86], Janssens et al. proposed Stochastic Outlier Selection (SOS) as a one-class classifier. SOS aims to create random graphs based on an affinity distribution to recognize frequently unlinked objects as outliers. Nevertheless, SOS could avoid the cost of computing graph sampling as the probability of a node being disconnected can be computed in closed-form multiplying the inverses of the probabilities of each node. SOS has shown good classification results in several applied contexts [86,87,90].

3.4. Neural networks

A Neural Network is a set of numeric methods endeavoring to recognize fundamental relationships in a set of data by a process inspired by the human brain functioning [91]. Neural Networks mean to frameworks of artificial neurons that can adjust to changing input, so the network generates the most precise

Table 2
Computational complexities reported by authors in their work.

Category	Name	Complexity
Linear	AAD LMDD	$O(n)$
	IQR LMDD	$O(n)$
	MCD	–
	ocSVM	Convex quadratic programming
	PCA	–
	VAR LMDD	$O(n)$
Proximity	Avg KNN	$O(n^2)$
	CBLOF	$O(n)$
	COF	**First step:** $O(n \cdot k)$ for low-dimensional data, $O(n \cdot log(n) \cdot k)$ for medium dimensional data, $O(n^2 \cdot k)$ for extremely high-dimensional data. **Second step:** $O(n)$
	HBOS	$O(n)$ in case of fixed bin width, $O(n \cdot log(n))$ using dynamic bin widths
	Largest KNN	$O(n^2)$
	LOF	**First step:** $O(n \cdot k)$ for low-dimensional data, $O(n \cdot log(n) \cdot k)$ for medium dimensional data, $O(n^2 \cdot k)$ for extremely high-dimensional data. **Second step:** O(n)
	Med KNN	$O(n^2)$
	SOD	$O(d \cdot n^2)$
Probabilistic	ABOD	$O(n^3)$
	Fast ABOD	$O(n^2 + n \cdot k^2)$
	SOS	–
Neural networks	MO-GAAL	–
	SO-GAAL	–
	VAE	–
Ensembles	AVE Bagging	–
	BRM	$O(n^2)$
	XGBOD	–
	ISOF	$O(n)$
	LODA	$O(n)$
	LSCP	$O(n \cdot d + n \cdot log(n) + s)$
	MAX Bagging	–
	OCKRA	–
	XGBOD	–

– The authors of the original article do not report the computational cost.

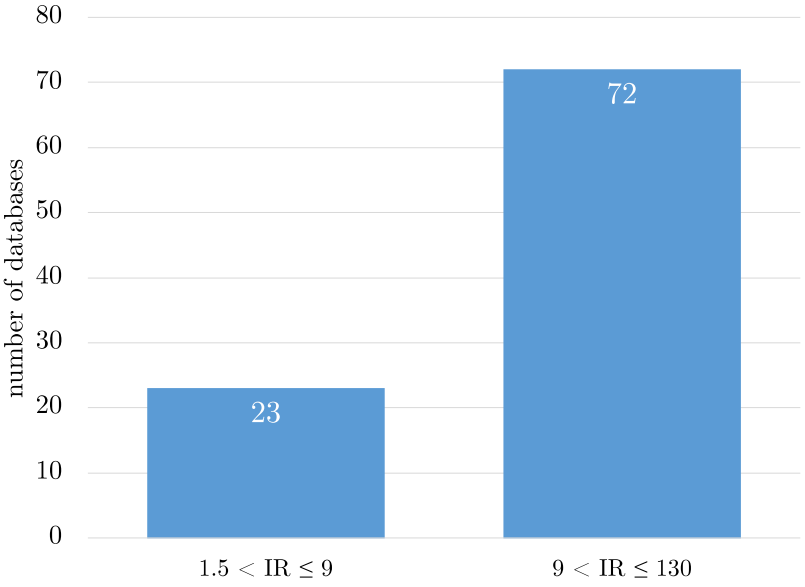


Fig. 3. The number of databases used in our experiments regarding the class imbalance ratio.

possible result. The concept of Neural Networks is quickly gaining popularity in the development of trading systems, and it has been useful for detecting anomalies too [92].

A Neural Network-based autoencoder is modeled as a function $f(x) = x$ that learns an estimation to the identity function, to output $\hat{x}$ that is similar to x. A Neural Network-based autoencoder uses the recreation error as the anomaly score [93,94].

VAE is a variant of autoencoder proposed by Kingma and Welling, which uses the L2 loss to compute the difference among recreated and original functions [96]. VAE's estimation uses the recreation error as the outlier score, showing good classification results on several real-world applications besides anomaly detection [90].

Single-Objective Generative Adversarial Active Learning (SO-GAAL) is a OCC and SSAD classifier proposed for anomaly

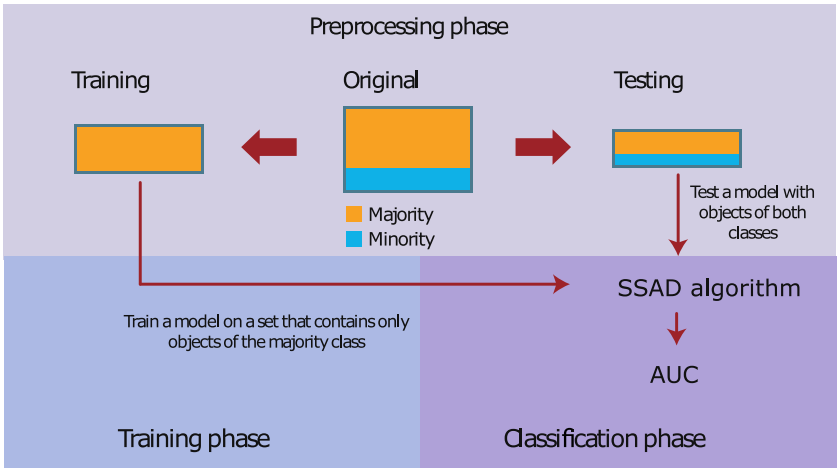


Fig. 4. Methodology employed in the experiments for each data fold.

detection [20,97]. SO-GAAL generates possible outliers based on the mini–max game by using two principal components (generator and discriminator) of a GAN. Furthermore, to avoid producing falling into the mode collapsing problem, the stop node of SO-GAAL should offer sufficiently valuable information. The same authors extended the network structure of SO-GAAL from a single generator to multiple generators with different objectives (MO-GAAL), which can produce an efficient reference distribution for the whole database [20].

3.5. Ensembles

Ensembles are classifiers taking advantage of the classification results of several individual classifiers. Diversity is the most crucial property of classifier ensembles. The main reason is that a collection of nearly identical classifiers cannot outperform any of their components [98,99].

Isolation Forest (a.k.a. iForest) was proposed by Liu et al. [100] as a OCC and SSAD classifier for outlier detection. iForest is an ensemble of several decision trees that generates predictions by feeding the input to internal trees and summarizing their outputs. iForest has shown better classification results than several state-of-the-art SSAD classifiers designed for anomaly detection [101].

An ensemble classifier called XGBOD (Extreme Gradient Boosting Outlier Detection) was proposed by Zhao and Hryniewicki. XGBOD was designed to enhance the detection of anomalies from individual observations in numerous collections. XGBOD combines the advantages of supervised and unsupervised classifiers by making a fusion approach that extracts the best information from them for anomaly detection. XGBOD mixes several unsupervised classifiers to generate a better representation from the data that enhances the classification result of a single semi-supervised classifier [28].

The SSAD classifiers mentioned above have shown good classification results in several contexts. Consequently, all of them – jointly with their variants – will be tested in our experiments for providing an up-to-date SSAD classifiers comparison.

Since computational complexity is a significant issue in real applications, for each method, we report the time complexity if it is reported by the authors of the algorithms. Table 2 shows the reviewed anomaly detection methods together with their computational complexities.

4. Empirical evaluation: Setup and findings

In this section, we will present the setup (e.g., the characteristics of the tested databases, the validation procedure executed, the tested SSAD algorithms, and parameter settings) for our evaluation (see Section 4.1) and the findings (see Section 4.2).

4.1. Experimental setup

The scarcity of benchmark databases for anomaly detection is a well-known problem [102]. For example, researchers such as those of [17,18,21] generally used multi-class databases and transformed them into anomaly evaluation databases. Some researchers [18,32,103] selected one of the classes as the target class (usually the majority class) and labeled the remaining classes as the positive class. In this way, a SSAD algorithm trains on a database that only has objects from the negative class, and then it is tested with objects from all the problem's classes.

In our experiments, we used 95 benchmark imbalanced databases (taken from the KEEL dataset repository[3] [104]) to evaluate the performance of the tested SSAD algorithms. Table 3 shows for each database their name (*Name*), the number of objects (*#Obj*), the number of features (*#Attr*), and the class imbalance ratio *(IR)*. The Imbalance Ratio (IR) indicates the fraction between the number of objects in the negative and positive classes [105]. Consequently, a higher IR value indicates more imbalance between the problem's classes, which means that the number of objects in the positive class is low compared to the negative class. We selected databases representing two-class problems and an IR greater than 1.5. The number of databases regarding the class imbalance ratio is described in Fig. 3.

Every database described in Table 3 was divided into five databases to perform 5-fold cross-validation (5-FCV) [106]. Then, for simulating a real anomaly detection situation, each training database only contains objects belonging to the negative class, and each testing database contains both negative and positive objects. As a result, all the selected SSAD algorithms will be trained with objects belonging to the negative class, and they will be tested with objects of both positive and negative classes. The overall methodology employed is outlined in Fig. 4.

Our study aims to compare several SSAD algorithms on many benchmark databases. Therefore, we selected 29 well-known and recent SSAD algorithms to evaluate their performance over the

[3] http://sci2s.ugr.es/keel/datasets.php.

Table 3
Characteristics of the databases used in our experiments. #Obj, #Feat, and IR stand for: number of objects, number of features, and class imbalance ratio, respectively.

Name	#Obj	#Feat	IR	Name	#Obj	#Feat	IR
glass1	214	9	1.82	ecoli0146vs5	280	6	13.00
ecoli0vs1	220	7	1.86	shuttlec0vsc4	1829	9	13.87
wisconsin	683	9	1.86	yeast1vs7	459	7	14.30
pima	768	8	1.87	glass4	214	9	15.46
iris0	150	4	2.00	ecoli4	336	7	15.80
glass0	214	9	2.06	pageblocks13vs4	472	10	15.86
yeast1	1484	8	2.46	abalone9vs18	731	8	16.40
haberman	306	3	2.78	dermatology6	358	34	16.90
vehicle2	846	18	2.88	zoo3	101	16	19.20
vehicle1	846	18	2.90	glass016vs5	184	9	19.44
vehicle3	846	18	2.99	shuttlec2vsc4	129	9	20.50
glass0123vs456	214	9	3.20	shuttle6vs23	230	9	22.00
vehicle0	846	18	3.25	yeast1458vs7	693	8	22.10
ecoli1	336	7	3.36	glass5	214	9	22.78
newthyroid1	215	5	5.14	yeast2vs8	482	8	23.10
newthyroid2	215	5	5.14	lymphography*	148	18	23.67
ecoli2	336	7	5.46	flareF	1066	11	23.79
segment0	2308	19	6.02	cargood	1728	6	24.04
glass6	214	9	6.38	carvgood	1728	6	25.58
yeast3	1484	8	8.10	krvskzeroonevsdraw	2901	6	26.63
ecoli3	336	7	8.60	krvskonevsfifteen	2244	6	27.77
pageblocks0	5472	10	8.79	yeast4	1484	8	28.10
ecoli034vs5	200	7	9.00	winequalityred4	1599	11	29.17
yeast2vs4	514	8	9.08	poker9vs7	244	10	29.50
ecoli067vs35	222	7	9.09	yeast1289vs7	947	8	30.57
ecoli0234vs5	202	7	9.10	abalone3vs11	502	8	32.47
glass015vs2	172	9	9.12	winequalitywhite9vs4	168	11	32.60
yeast0359vs78	506	8	9.12	yeast5	1484	8	32.73
yeast0256vs3789	1004	8	9.14	krvskthreevseleven	2935	6	35.23
yeast02579vs368	1004	8	9.14	winequalityred8vs6	656	11	35.44
ecoli046vs5	203	6	9.15	ecoli0137vs26	281	7	39.14
ecoli01vs235	244	7	9.17	abalone17vs78910	2338	8	39.31
ecoli0267vs35	224	7	9.18	abalone21vs8	581	8	40.50
glass04vs5	92	9	9.22	yeast6	1484	8	41.40
ecoli0346vs5	205	7	9.25	winequalitywhite3vs7	900	11	44.00
ecoli0347vs56	257	7	9.28	winequalityred8vs67	855	11	46.50
yeast05679vs4	528	8	9.35	abalone19vs10111213	1622	8	49.69
vowel0	988	13	9.98	krvskzerovseight	1460	6	53.07
ecoli067vs5	220	6	10.00	winequalitywhite39vs5	1482	11	58.28
glass016vs2	192	9	10.29	poker89vs6	1485	10	58.40
ecoli0147vs2356	336	7	10.59	shuttle2vs5	3316	9	66.67
led7digit02456789vs1	443	7	10.97	winequalityred3vs5	691	11	68.10
ecoli01vs5	240	6	11.00	abalone20vs8910	1916	8	72.69
glass06vs5	108	9	11.00	krvskzerovsfifteen	2193	6	80.22
glass0146vs2	205	9	11.06	poker89vs5	2075	10	82.00
glass2	214	9	11.59	poker8vs6	1477	10	85.88
ecoli0147vs56	332	6	12.28	abalone19	4174	8	129.44
cleveland0vs4	177	13	12.62				

selected 95 benchmark imbalanced databases described in Table 3. All the selected classifiers were executed using the parameters recommended by their authors, which are shown in Tables 4–5. It is important to highlight that some of the selected SSAD algorithms, like ocSVM, cannot deal with nominal data. Consequently, we use OneHotEncoder[4] as an encoding method for all nominal features. OneHotEncoder was executed by using the scikit-learn library of Python [107], which has shown good results in different contexts.

We also used the Area Under the Curve (AUC) of the Receiver Operating Characteristic curve to assess the performance of the classifiers tested in our experiments. We chose this metric because it is frequently used to evaluate SSAD algorithms, it also provides a summary of the overall performance of the classifiers, and it is insensitive to the distribution of classes in the database [116]. As a perfect result, we would have an AUC equal to 1, but in real-life situations, a classifier usually cannot wholly separate objects of the negative class from those of the positive class. AUC values above 0.5 are preferable since a value equal to 0.5 indicates that the classifier's performance is random guessing and lower than 0.5 means that the classifier's classification scheme should be inverted. We computed the AUC using the implementation provided by the scikit-learn library of Python [107].

We applied the Friedman's test [117–119] to see whether there is a significant statistical difference among the results obtained by the tested classifiers. Given a set of k classifiers (where $k \geq 2$) and N databases, the Friedman's test ranks the classifiers according to the result obtained in each database as follows:

1. For each database i, Friedman's test assigns values from 1 (to the best result) to k (to the worst result). In the case of ties, the average ranks are assigned. The ranks are denoted as r_i^j (where $1 \leq j \leq k$ and $1 \leq i \leq N$).
2. To obtain the final rank for each tested classifier j, the Friedman test averages the ranks obtained in all databases, $R_j = \frac{1}{N} \sum_i r_i^j$.

[4] https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html.

Table 4
Parameter specification for the SSAD algorithms tested in our experimentation (Part I).

Ref.	Abbr.	Full name	Parameters
[108]	AAD LMDD	Average Absolute Deviation Linear Method for Deviation-based Outlier Detection	Dissimilarity Measure (DM): Average absolute deviation
[25]	ABOD	Angle-Based Outlier Detection	Number of Neighbors (K): 25; Method (M): Normal
[109]	AVE Bagging	Average Bagging	Base Classifier (BC): LOF; Number of Classifiers (N): 10; Combination Method (C): Average
[110]	Avg KNN	Average k-Nearest Neighbors	K: 5; M: Mean; Distance Metric (D): Minkowski
[26]	BRM	Bagging-Random Miner	N: 100; Fraction to Bootstrap: 20%
[80]	CBLOF	Clustering-Based Local Outlier Factor	Number of Clusters of K-means (N_Clusters): 8; Alpha (α): 0.9; Beta (β): 5
[79]	COF	Connectivity-Based Outlier Factor	K: 20
[111]	eForest	Encoder Forest	Split Evaluation: Random; Decision Trees: 1000; Tree Depth (TD): Maximum
[25]	Fast ABOD	Fast Angle-Based Outlier Detection	K: 5; M: Fast
[112]	HBOS	Histogram-based Outlier Score	Number of Bins (N_Bins): 10; α: 0.1; Tolerance (T): 0.5
[108]	IQR LMDD	Interquartile Range Linear Method for Deviation-based Outlier Detection	DM: Interquartile Range
[53]	ISOF	Isolation Forest	N: 100; Number of Samples: Auto
[15]	Largest KNN	Largest k-Nearest Neighbors	K: 5; M: Largest; DM: Minkowski
[113]	LODA	Lightweight On-line Detector of Anomalies	N_Bins: 10; Number of Random Cuts: 100
[14]	LOF	Identifying Density-Based Local Outliers	K:20, M:auto, Leaf Size (LS): 30, D: Minkowski
[114]	LSCP	Locally Selective Combination in Parallel outlier ensembles	Local Region Size (LRS):30, Bins (B): 10

Table 5
Parameter specification for the SSAD algorithms tested in our experimentation (Part II).

Ref.	Abbr.	Full name	Parameters
[109]	MAX Bagging	Maximum Bagging	BC: LOF; N: 5; C: Maximum
[79]	MCD	Minimum Covariance Determinant	Store Precision: True; Assume Centered: False
[15]	Med KNN	Median k-Nearest Neighbors	K: 5; M: Median; DM: Minkowski
[20]	MO-GAAL	Multiple-Objective Generative Adversarial Active Learning	Number of Sub-generators: 10; Number of Epochs (E): 20; Discriminator Learning Rate (DLR): 0.01; Generator Learning Rate (GLR): 0.0001; SGD Decay (*Dec*): 1e−6; SGD Momentum (P): 0.9
[19]	OCKRA	One-Class K-means with Randomly-projected features Algorithm	N: 100; N_Clusters: 10
[13]	ocSVM	One-Class Support Vector Machines	Kernel Type: Radial basis function; Degree: 3; Gamma (γ): auto; T: 0.001; Nu: 0.5
[57]	PCA	Principal Component Analysis	Number of Components: min(n_samples, n_feat); SVD Solver: Auto; T: 0.0, Standardization: True; Weighted: True
[20]	SO-GAAL	Single-Objective Generative Adversarial Active Learning	E: 20; DLR: 0.01; GLR: 0.0001; *Dec*: 1e−6; P: 0.9
[115]	SOD	Subspace Outlier Detection	K: 20; Number of Shared Nearest Neighbors: 10; α: 0.8
[86]	SOS	Stochastic Outlier Selection	Perplexity: 4.5; DM: Euclidean; EPS: 1e−5
[95]	VAE	Variational AutoEncoder	Hidden Activation: ReLU; Output Activation: Sigmoid; Loss: Mean squared error; Optimizer: Adam; E: 100; Batch Size: 32; Dropout Rate: 0.2; L2 Regularizer: 0.1; Validation Size: 0.1
[108]	VAR LMDD	Variance Linear Method for Deviation-based Outlier Detection	DM: Variance
[28]	XGBOD	Extreme Boosting Based Outlier Detection	TD: 3; Learning Rate (LR): 0.1; Number of Boosted Trees: 100; Objective Function: Binary logistic; Booster: Gbtree; γ: 0; Bias: 0.5

3. Finally, Friedman's test is computed by using Eq. (1), which is distributed according to a X^2 distribution.

$$F_f = \frac{12N}{k(k+1)}\left[\sum_j R_j^2 - \frac{k(k+1)^2}{4}\right] \quad (1)$$

Friedman's test is suitable for validating or rejecting the null-hypothesis. Commonly, in Machine Learning, the null-hypothesis aims to corroborate if the results obtained by all the tested classifiers, on all databases, are similar statistically. For that, a significance level (α) should be fixed, which helps to decide if the null-hypothesis is rejected or accepted. In our experimental setup, we used the most common significance level, $\alpha = 0.05$ [117–119].

It is important to notice that Friedman's test only identifies whether there is or not a statistical difference in the compared classifiers. Therefore, if the null-hypothesis is rejected, we can proceed with a post-hoc test. This type of test will help us to find which of the tested classifiers are different among them (n x n comparison). We selected the Shaffer post-hoc procedure, as suggested in [119,120], because it offers good results and requires less computational time when comparing many classifiers.

Table 6
Average AUC, the standard deviation (STD), and average rankings (based on the Friedman's test) and the p-values (based on the Shaffer's procedure) for all the classifiers using all the tested databases. These results are ordered according to the Friedman's ranking.

Classifier	Average AUC	STD	Ranking	Adjusted p-value
BRM	0.8347	0.1442	3.9684	–
OCKRA	0.8110	0.1491	5.0000	0.4037
ISOF	0.7785	0.1410	6.2211	0.0683
ocSVM	0.7616	0.1493	7.4526	0.0048
eForest	0.7439	0.1524	9.0053	0.0001
Avg KNN	0.7261	0.1532	10.2579	0.0000
Med KNN	0.7185	0.1566	11.0211	0.0000
Largest KNN	0.7094	0.1549	11.7947	0.0000
LOF	0.6984	0.1511	12.7947	0.0000
LSCP	0.6974	0.1331	12.7474	0.0000
MAX Bagging	0.6957	0.1402	12.8842	0.0000
AVE Bagging	0.6946	0.1466	13.1158	0.0000
CBLOF	0.6877	0.1522	14.5526	0.0000
ABOD	0.6853	0.1615	13.1421	0.0000
PCA	0.6841	0.1441	14.0684	0.0000
HBOS	0.6780	0.1419	14.3053	0.0000
VAE	0.6767	0.1459	14.6211	0.0000
Fast ABOD	0.6766	0.1670	13.8000	0.0000
MCD	0.6519	0.1404	14.8526	0.0000
LODA	0.5956	0.0932	18.5000	0.0000
SOD	0.5867	0.0752	17.4211	0.0000
COF	0.5678	0.0777	19.0737	0.0000
MO-GAAL	0.5603	0.0849	20.3737	0.0000
SO-GAAL	0.5590	0.0782	21.0263	0.0000
AAD LMDD	0.5332	0.0685	23.3842	0.0000
SOS	0.5326	0.0332	21.7526	0.0000
VAR LMDD	0.5299	0.0538	23.7737	0.0000
IQR LMDD	0.5133	0.0313	26.0263	0.0000
XGBOD	0.5046	0.0445	28.0632	0.0000

4.2. Results and discussion

In Fig. 5, we depict the averaged AUC of each classifier on all the 95 benchmark imbalanced databases using a boxplot [121]. The boxplot (also called box-and-whisker plot) displays the distribution of the data based on a summary of five values (minimum, first quartile, median, third quartile, and maximum) [121].

From Fig. 5, we can see that most of the tested SSAD algorithms have a median value greater than 0.5, excepting XGBOD. From these results, we can notice that the BRM's boxplot shows the best values regarding the remaining tested classifiers. Notice that the average AUC value obtained by BRM is 0.83, and its third quartile, on the boxplot, shows that the results are above 0.95. In general, from Fig. 5, we can see that BRM and OCKRA perform the best, followed by ISOF and ocSVM. On the other hand, the first-five classifiers showed in Fig. 5 (XGBOD, IQR LMDD, VAR LMDD, SOS, and AAD LMDD) have average AUC value between 0.5 and 0.55, indicating poor classification results.

Table 6 shows the average AUC, standard deviation (SD), average ranking according to the Friedman's test, and the adjusted p-value of the Shaffer's procedure for each tested SSAD algorithms. We have sorted Table 6 based on the obtained Friedman's ranking values, and we have indicated with a thin-horizontal line the point after which there is a statistically significant difference against the best result obtained by BRM (Adjusted p-value ≤ 0.05). According to Table 6, BRM achieved the best position, followed by OCKRA and ISOF. Adjusted p-values show that BRM significantly outperformed the results of 26 classifiers (from ocSVM to XGBOD). However, there was no statistically significant difference in the results obtained by BRM, OCKRA, and ISOF. Therefore, to compare BRM against OCKRA and ISOF, we executed a pairwise comparison using the Wilcoxon signed-rank test [122]. The Wilcoxon test presents the sums of the ranks of the two models being compared and determines whether the difference between these measurements is due to randomness or not.

Table 7
Results of Wilcoxon signed-ranks test comparing BRM against OCKRA and ISOF, using all the tested databases.

Comparison	R^+	R^-	Hypothesis ($\alpha = 0.05$)	p-value
BRM vs. OCKRA	3074	1112	Rejected	$< .00001$
BRM vs. ISOF	3451	1109	Rejected	$< .00001$

Table 7 shows the comparison of BRM against OCKRA and ISOF, the sum of the ranks for the databases where BRM improved OCKRA or ISOF (R^+), the sum of the ranks for the opposite (R^-), the result of the null hypothesis (Hypothesis), and the p-value obtained in the Wilcoxon signed-rank test. From Table 7, we can see that the p-value of the test is < 0.00001, which is lower than the significance level of $\alpha = 0.05$. We can confirm that the BRM classifier significantly outperformed the AUC results of OCKRA and ISOF.

Finally, from the results shown in our experiments, we can conclude that the BRM classifier achieved an outstanding performance compared to 29 state-of-the-art algorithms using 95 different benchmark imbalanced databases.

4.2.1. Meta-analysis study

A database is a collection of objects, also called instances, records, or samples. In addition, it is described as a set of features, which are the columns of the data (also referred to as attributes or variables). Features can be of different types, such as numeric, nominal, or mixed. The values of a numerical feature are numbers (discrete and continuous), nominal feature values are data that is used for naming or labeling, without any quantitative value, e.g., country names, hair colors, race, gender, etc. [8], and mixed databases combine both types of data.

In this section, we provide an in-depth study of the tested SSAD algorithms regarding the features of the different tested databases, such as the total number of objects, type of features, and IR. For doing that, we created a database collecting the features of the databases used in our experimental setup. Afterward, we executed k-Means [123], Single Linkage (hierarchical) [124], and expectation–maximization [125] as clustering algorithms for finding clusters ranging from two to five. After that, we executed silhouettes as a validity index for selecting the best clusters [126]. We selected these clustering and validity algorithms because they have shown good clustering results on several problems [127, 128]. Table 8 shows the description of the clusters found in the databases used for our experiments.

For a better illustration of the size distribution of clusters found, Fig. 6 shows the number of databases regarding the number of objects, the number of features, the data type, and the IR. From this figure, we can see that the number of databases regarding the number of objects is slightly imbalanced. However, notice that the clusters obtained regarding the number of features and the feature type present more imbalanced data than those found regarding the number of objects and IR.

For this meta-analysis, we will analyze the top-10 ranked SSAD algorithms shown in Table 6 and Fig. 5. First, we will discuss the obtained results regarding the number of objects. After, we will analyze the results but regarding the number of features. Next, we will comment on the results taking int account the feature types and, finally, regarding the IR.

Analyzing the performance of the SSAD algorithms regarding the clusters obtained from the number of objects, we can see that in Cluster 1 (Objects < 443), there is a remarkable improvement for the top-10 analyzed SSAD algorithms, being the most accurate

Table 8
Clusters description.

Databases' features	Cluster1	Cluster2	Cluster3
#Objects	*Objects* $<$ 443	433 $\geq$ *Objects* $<$ 1, 460	*Objects* $\geq$ 1460
#Features	*Features* $<$ 13	*Features* $\geq$ 13	–
Type	*Numeric*	*Nominal*	*Mixed*
IR	*IR* $<$ 19.20	19.20 $\geq$ *IR* $<$ 39.14	*IR* $\geq$ 39.14

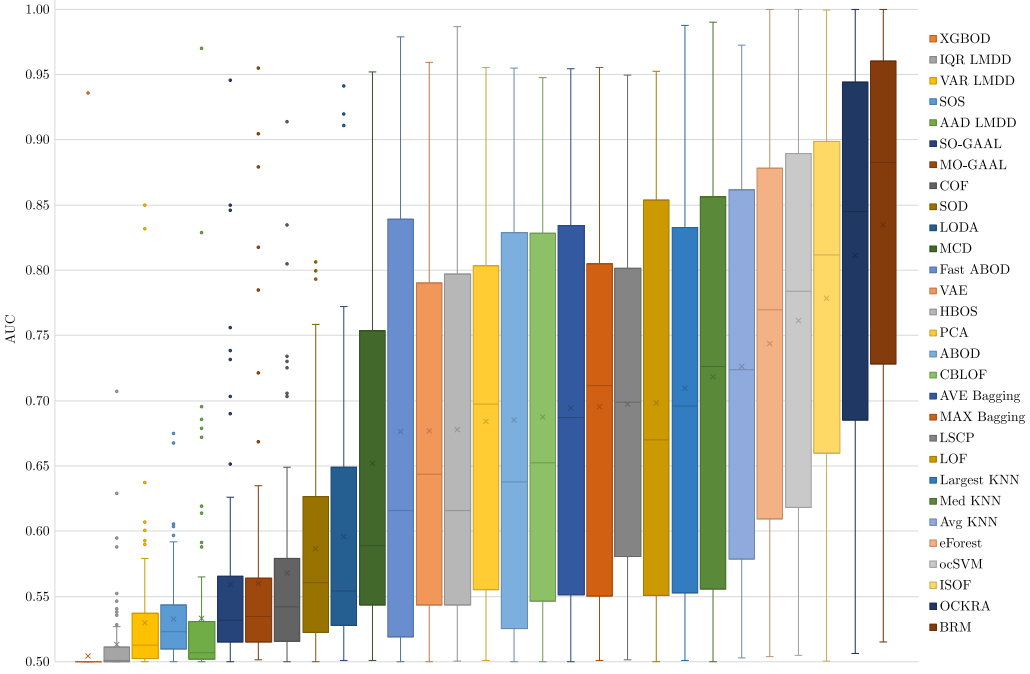


Fig. 5. Boxplots showing the performance (in terms of AUC) of each tested SSAD algorithms.

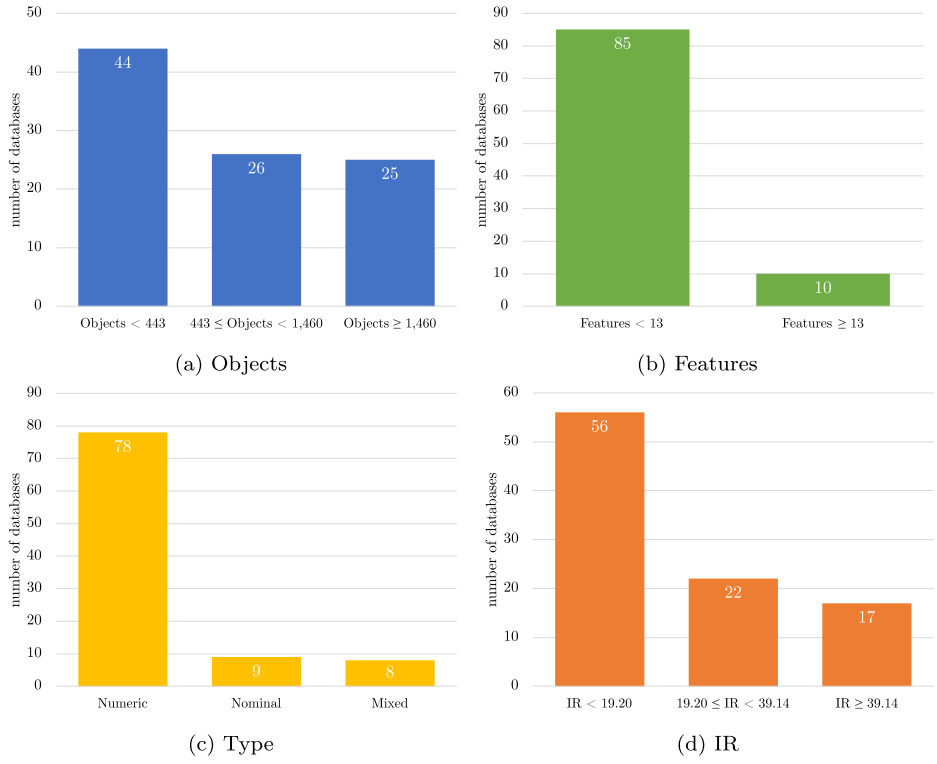
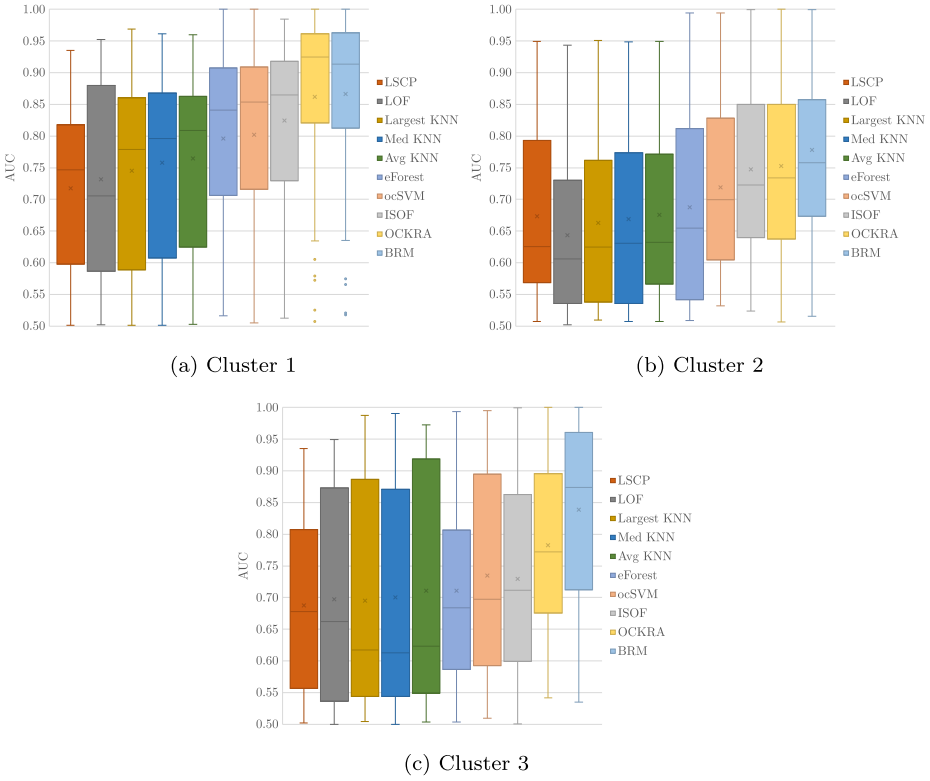


Fig. 6. The size distribution of the clusters found regarding the number of objects and features, type of features, and IR of the tested databases.



(a) Cluster 1 (b) Cluster 2

(c) Cluster 3

Fig. 7. Boxplot of the top 10 SSAD algorithms regarding the clusters found according to the number of objects.

Table 9
Average AUC, for the top 10 SSAD algorithms regarding the clusters found regarding the number of objects. These results are sorted according to the average AUC presented in Table 6 (Global AUC).

Classifier	Global AUC	Average AUC by cluster		
		Cluster 1	Cluster 2	Cluster 3
BRM	0.8347	0.8661	0.7781	0.8383
OCKRA	0.8110	0.8619	0.7525	0.7824
ISOF	0.7785	0.8247	0.7473	0.7297
ocSVM	0.7616	0.8023	0.7189	0.7346
eForest	0.7439	0.7963	0.6872	0.7105
Avg KNN	0.7261	0.7646	0.6754	0.7109
Med KNN	0.7185	0.7581	0.6689	0.7005
Largest KNN	0.7094	0.7453	0.6629	0.6948
LOF	0.6984	0.7317	0.6433	0.6971
LSCP	0.6974	0.7171	0.6732	0.6878

Table 10
Average AUC, for the top 10 SSAD algorithms regarding the clusters found in the number of features. These results are ordered according to the average AUC presented in Table 6 (Global AUC).

Classifier	Global AUC	Average AUC by cluster	
		Cluster 1	Cluster 2
BRM	0.8347	0.8301	0.8739
OCKRA	0.8110	0.8075	0.8413
ISOF	0.7785	0.7812	0.7555
ocSVM	0.7616	0.7709	0.6833
eForest	0.7439	0.7433	0.7492
Avg KNN	0.7261	0.7286	0.7046
Med KNN	0.7185	0.7236	0.6754
Largest KNN	0.7094	0.7150	0.6623
LOF	0.6984	0.7003	0.6823
LSCP	0.6974	0.7020	0.6580

BRM and OCKRA. Notice that in Table 9, BRM and OCKRA achieved a very similar average AUC, and in Fig. 7a, their boxplots are almost at the same level, and a positive skewness is visible for most of the boxplots.

Besides, for Cluster 2 ($443 \leq$ Objects < 1460) and Cluster 3 (Objects ≥ 1460) all the algorithms decreased their performance. The most remarkable difference is the ISOF's performance with an average AUC equal to 0.7297 from 0.7785 obtained in the Global AUC shown in Table 6.

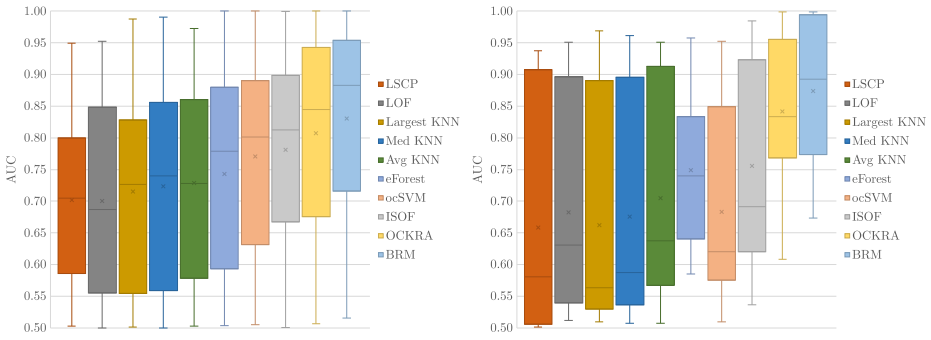
Comparing Fig. 7 and Table 9 against Fig. 5 and Table 6, we can notice that the top-10 SSAD algorithms improve their results when the number of objects in the databases is lower than 443. Besides, we can see that the analyzed classifiers perform worse when the number of objects in the databases is greater than 443.

According to the number of features, Table 10 shows that there are no differences between the results obtained in Cluster 1 ($13 <$ Features) and those obtained in Table 6. These results can be explained by the fact that 90% of the databases are concentrated in this cluster. On the other hand, in Cluster 2 (Features ≥ 13), we notice an improvement in the average AUC of BRM and OCKRA (0.0392 and 0.0303, respectively). It is important to highlight that ocSVM fell to sixth place with an average AUC of 0.6833 from 0.7616 obtained in Table 6.

Also, as shown in Fig. 8b, most of the boxplots have negative skewness. The median AUC value of LSCP, Largest KNN, and Med KNN is less than 0.60. Nevertheless, note that the third quartile of BRM and OCKRA are significantly higher than the others. The third quartile of BRM is almost 1.

Another analysis we carried out was regarding the type of feature (Numeric, Nominal, and Mixed), which are shown in Table 11 and Fig. 9. In Cluster 1 (Type = Numeric), the results remained very similar to the results obtained in Table 6 and Fig. 5. As in the previous case, more than four-fifths of the databases are concentrated in this cluster; consequently, the results shown in Table 11 have no notable differences against those shown in Table 6 regarding the same top-10 analyzed classifiers in Table 11.

On the one hand, in Cluster 2 (Type = Nominal), ocSVM and Avg KNN improves their performance and have the second and



(a) Cluster 1 (b) Cluster 2

Fig. 8. Boxplot of the top 10 SSAD algorithms regarding the clusters found according to the number of features.

Table 11
Average AUC, for the top 10 SSAD algorithms regarding the clusters found regarding the type of features. These results are ordered according to the average AUC presented in Table 6 (Global AUC).

Classifier	Global AUC	Average AUC by cluster		
		Cluster 1	Cluster 2	Cluster 3
BRM	0.8347	0.8288	0.8668	0.8563
OCKRA	0.8110	0.8103	0.7831	0.8493
ISOF	0.7785	0.7804	0.7915	0.7459
ocSVM	0.7616	0.7583	0.8261	0.7216
eForest	0.7439	0.7490	0.7563	0.6799
Avg KNN	0.7261	0.7124	0.8010	0.7748
Med KNN	0.7185	0.7079	0.7622	0.7733
Largest KNN	0.7094	0.6982	0.7495	0.7736
LOF	0.6984	0.6904	0.7239	0.7480
LSCP	0.6974	0.6886	0.7814	0.6889

Table 12
Average AUC, for the top 10 SSAD algorithms regarding the clusters found regarding the IR. These results are ordered according to the average AUC presented in Table 6 (Global AUC).

Classifier	Global AUC	Average AUC by cluster		
		Cluster 1	Cluster 2	Cluster 3
BRM	0.8347	0.8355	0.8071	0.8679
OCKRA	0.8110	0.8272	0.7795	0.7986
ISOF	0.7785	0.7917	0.7867	0.7246
ocSVM	0.7616	0.7713	0.7760	0.7112
eForest	0.7439	0.7685	0.6884	0.7344
Avg KNN	0.7261	0.7246	0.7247	0.7327
Med KNN	0.7185	0.7207	0.7103	0.7220
Largest KNN	0.7094	0.7089	0.7003	0.7232
LOF	0.6984	0.6961	0.7015	0.7022
LSCP	0.6974	0.6990	0.6948	0.6955

third best average AUC, respectively. OCKRA dropped its performance to 0.7831 from 0.8110. Furthermore, while LSCP, Med KNN, avg KNN, OCSVM, and BRM significantly improved their median value, OCKRA and ISOF dropped their median value, as shown in Fig. 9b. Notice that the third-quartile of BRM is very close to 1. On the other hand, Fig. 9c shows the boxplots for Cluster 3 (Type = Mixed). From this figure and Table 11, we can see that ISOF, eForest, and ocSVM decreased their median value less than 0.7. While LSCP, ocSVM, and ISOF have a remarkable negative skewness for their boxplots.

Finally, we will discuss what happens when the classifiers deal with different class imbalance levels. The imbalance ratio (IR) is related to the ratio between objects belonging to the negative class and those belonging to the positive class. Usually, anomaly detection problems have imbalanced classes. Therefore, those classifiers improving their performance on imbalanced databases are desirable.

On the one hand, as shown in Table 12, in Cluster 1 (IR $<$ 19.29), the average AUC of the analyzed SSAD algorithms are slightly better compared to the overall AUC in Table 6. Notice that BRM and OCKRA obtained the best results, achieving an average AUC equal to 0.8355 and 0.8272. On the other hand, in Cluster 2 (19.20 $\leq$ IR $<$ 39.14), OCKRA drops its performance and has an average AUC equal to 0.7795. However, as we can see in Table 12 and Fig. 10b eForest has the worst performance among the top-10 analyzed classifiers, and it has achieved a poor average AUC of 0.6884. Besides, notice that in Cluster 3 (IR $\geq$ 39.14), there is a notable improvement of BRM over the rest of the analyzed classifiers, obtaining the highest AUC (0.8670), and the second-best is OCKRA with 0.7986 of AUC. For more detail, Fig. 10c shows that BRM has the smallest boxplot and the largest median value. Also, the third quartile is above 0.95.

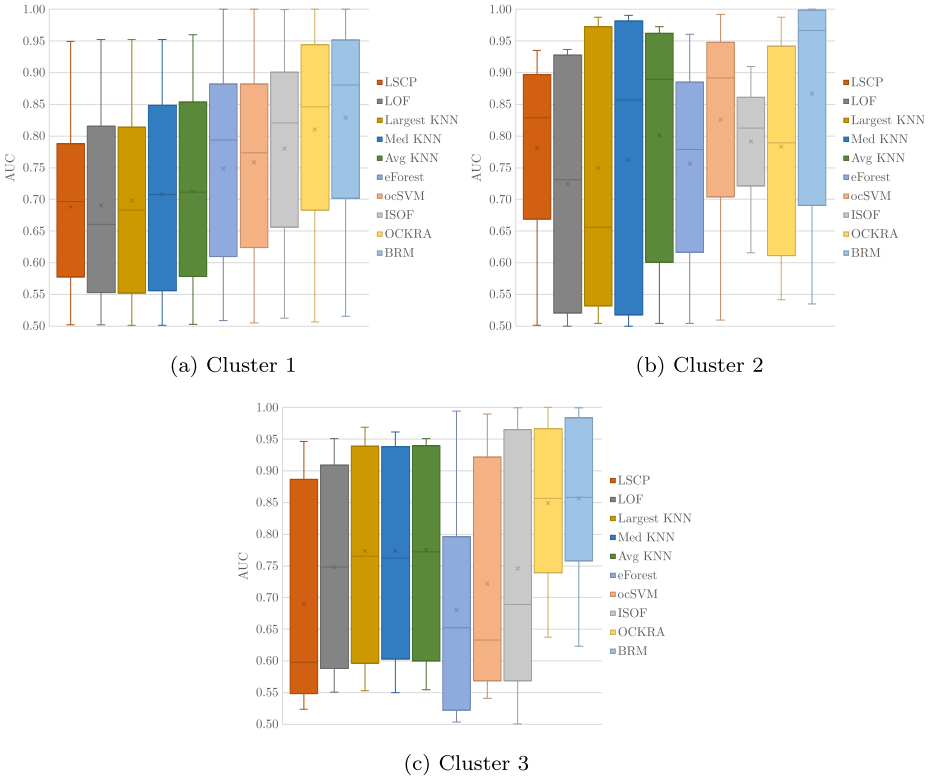
We can conclude from our study that BRM achieved the best result over the remaining 28 SSAD algorithms on 95 imbalanced databases. From our meta-analysis, we can conclude that BRM always obtained the best result on the different clusters regarding the total number of objects and features, type of features, and IR. Consequently, we can argue that BRM is a robust classifier obtaining good classification results on diverse anomaly detection problems. Also, we can conclude that OCKRA, ISOF, and ocSVM obtained a good performance overall AUC, but they achieved worse classification results when the databases contain the number of objects equal or greater than 1460, the type of feature is nominal, or a high-class imbalance ratio (IR $\geq$ 39.14).

5. Conclusion and future research

5.1. Summary

SSAD approaches play an increasingly important role in detecting anomalies, particularly in complex and interconnected systems/applications. However, it can be challenging for the research and practitioner communities to keep pace with the constant advances in the literature. This reinforces the importance of having ongoing empirical evaluations of state-or-the-art classifiers, for example, to inform the research and practitioner communities on which classifier(s) to choose for some specific scenario. Existing empirical evaluations generally focus on solving a specific problem; consequently, their experimental setups contain a small number of SSAD algorithms and databases for testing. Thus, it is necessary to systematically evaluate the performance of a more comprehensive set of SSAD algorithms on different databases.

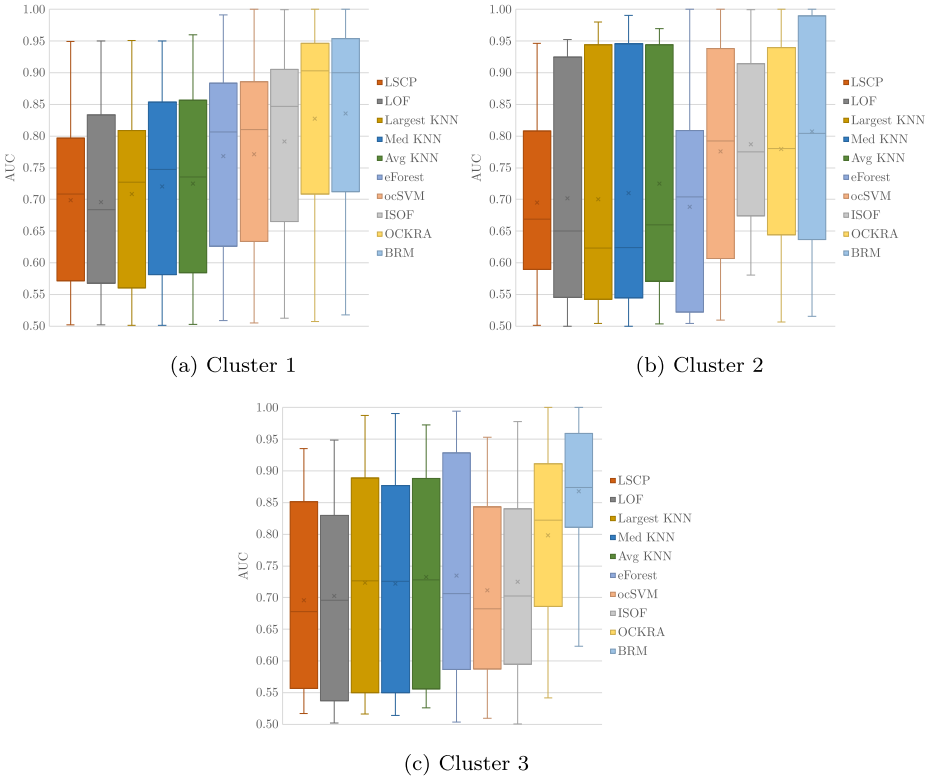
In this paper, we empirically evaluated the performance of 29 state-of-the-art SSAD algorithms (belonging to five broad categories) on 95 imbalanced benchmark databases, taken from the KEEL repository. Our studies showed that BRM, OCKRA, and



(a) Cluster 1 (b) Cluster 2

(c) Cluster 3

Fig. 9. Boxplot of the top 10 SSAD algorithms regarding the clusters found according to the type of features.



(a) Cluster 1 (b) Cluster 2

(c) Cluster 3

Fig. 10. Boxplot of the top 10 SSAD algorithms regarding the clusters found according to the IR.

Isolation Forest (iForest or ISOF) achieved the highest average AUC, and they also ranked highest according to Friedman's test. However, it is important to highlight that, based on Shaffer and Wilcoxon signed-rank tests, BRM significantly outperformed OCKRA and ISOF, corroborating that BRM obtained the best classification results with statistical differences.

Our meta-analysis showed that BRM obtained the best result on the different clusters, in terms of the total number of objects

and features, type of features, and IR. Also, our findings showed that BRM is a robust classifier obtaining good classification results on diverse anomaly detection problems. Besides, from our meta-analysis, we can conclude that OCKRA, ISOF, and ocSVM obtained good performance in terms of the overall AUC, but their classification performance degraded when the number of objects in the database is equal or greater than 1460, the type of feature is nominal, or the IR is equal or greater than 39.14.

5.2. Future research agenda

There are several potential extensions to this work. First, we need to provide an up-to-date comparison of SSAD algorithms regarding their training and classification time and their computational complexity. For several application contexts, it is necessary to know such parameters due to the implementation on devices with limited memory and CPU resources. In addition to the computation time, it is worth studying the performance using different algorithms of feature selection and parameter-tuning.

Due to limited anomalies, SSAD models are built only by using normal data. However, such models may lead to inaccurate results, particularly when the normal data contain noise or uncertainty. To address these limitations, future research should also include tests that examine the robustness of methods against noise in the evaluation database and features.

Classifier ensembles for SSAD, specifically using the boosting method, is a topic that is understudied (see Table 2). Boosting can generate a combined model with lower errors, optimizing the boosting's advantages and reducing the single model's errors. However, the most used ocSVM's kernels are Linear, Polynomial, Gaussian, or Sigmoidal. Hence, future research efforts should also explore the potential of other kernels, like Genetic Kernels or Weighted Linear Spectral Kernel [129], which have shown promising results in supervised problems.

In recent times, there has been growing interest in the creation of models based on eXplainable Artificial Intelligence (XAI) [130]. One approach based on XAI is Contrast Pattern Classification, which has shown better classification results than other popular state-of-the-art classifiers, and it provides an explanation of the results close to experts in the application area [131]. A promising idea is to develop contrast pattern-based classifiers for obtaining a one-class decision boundary built from an explanatory point of view.

Another potential future research direction is to explore the use of Bayesian Network in applications involving several normal instances, with only a few positive instances (a big data problem) [132]. The main idea is to identify some reliable/strong normal instances for creating a good borderline. The Bayesian Network approach works quickly, saving much time and estimating prior distribution with minimal user intervention.

CRediT authorship contribution statement

Miryam Elizabeth Villa-Pérez: Methodology, Formal analysis, Writing - original draft, Investigation, Writing - review & editing. **Miguel Á. Álvarez-Carmona:** Methodology, Formal analysis, Writing - original draft, Investigation, Writing - review & editing. **Octavio Loyola-González:** Methodology, Formal analysis, Writing - original draft, Investigation, Formal analysis, Software, Writing - review & editing, Supervision. **Miguel Angel Medina-Pérez:** Methodology, Formal analysis, Writing - original draft, Investigation, Formal analysis, Software, Writing - review & editing, Supervision. **Juan Carlos Velazco-Rossell:** Investigation, Formal analysis, Software. **Kim-Kwang Raymond Choo:** Methodology, Formal analysis, Writing - original draft, Investigation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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